ANSWER KEY & MARKING SCHEME SET A

Geography Chapter 5: **Minerals and Energy Resources** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** bauxite	Residual deposit
2	**(b)** mica	Chota Nagpur edge
3	**(c)** sedimentary rocks	Beds or layers
4	**(b)** thorium	Atomic energy
5	**(a)** Both A and R are true and R is the correct explanation of A	Valuable in electrical industry
6	**(c)** a 100 per cent export unit	Ore sent as slurry to Mangaluru
7	**(b)** anthracite	Coal types
8	**(d)** Neyveli, Tamil Nadu	Electricity generation
9	**(b)** Assam	Digboi
10	**(d)** Nagarcoil to Madurai	Wind energy

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A homogeneous, naturally occurring substance with a definable internal structure.	2
12	Ferrous minerals contain iron (iron ore, manganese, nickel, cobalt); non-ferrous minerals do not contain iron (copper, bauxite, lead, zinc, gold).	1+1
13	Coal mining in Jowai and Cherrapunjee (Meghalaya) done by family members in the form of a long narrow tunnel; declared illegal by the National Green Tribunal.	2
14	Alluvial deposits of minerals in sands of valley floors and the base of hills, containing minerals not corroded by water — gold, silver, tin, platinum.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	They occur in cracks, crevices, faults or joints; smaller occurrences are veins and larger ones lodes; formed when minerals in liquid/molten and gaseous forms are forced upward through cavities towards the surface where they cool and solidify; e.g. tin, copper, zinc, lead.	1+1+1
16	Minerals are finite and non-renewable; workable deposits are only one per cent of the earth's crust; formation is very slow; extraction from greater depths increases cost and lowers quality. Ways — use of improved technology to use low-grade ores; recycling of metals; using scrap metals and substitutes.	1½+1½
17	Conventional — used for a long time, mostly non-renewable and polluting: firewood, cattle dung cake, coal, petroleum, natural gas, electricity (hydel and thermal); non-conventional — renewable, eco-friendly, recent: solar, wind, tidal, geothermal, biogas and atomic energy.	3
	OR: Dust and noxious fumes cause pulmonary diseases; risk of collapsing mine roofs, inundation and fires in coal mines; water sources get contaminated; dumping of waste and slurry degrades land and soil; stream and river pollution; stricter safety and environmental laws needed.	3
18	India is a tropical country with enormous sunshine; photovoltaic technology converts sunlight directly into electricity; popular in rural and remote areas; big solar plants being established; reduces dependence on firewood and dung cakes; environmental conservation and more manure for agriculture.	3
	OR: Gas produced from decomposition of shrubs, farm waste, animal and human waste; higher thermal efficiency than kerosene, dung cake and charcoal; gobar gas plants give twin benefits of energy and improved manure; prevents loss of trees and manure due to burning.	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	Most abundant fossil fuel; Gondwana coal (a little over 200 million years) — metallurgical coal; Damodar valley (West Bengal–Jharkhand) — Jharia, Raniganj, Bokaro; Godavari, Mahanadi, Son and Wardha valleys; Tertiary coal (about 55 million years) — Meghalaya, Assam, Arunachal Pradesh, Nagaland; lignite at Neyveli (TN); coal is bulky, so heavy industries and thermal power stations are located on or near coalfields.	5
	OR: Odisha–Jharkhand belt — Badampahar (Mayurbhanj, Kendujhar), Gua and Noamundi (Singhbhum); Durg–Bastar–Chandrapur belt (Chhattisgarh–Maharashtra) — Bailadila hills, 14 deposits of super high grade hematite, exported to Japan and South Korea via Visakhapatnam; Ballari–Chitradurga–Chikkamagaluru–Tumakuru belt (Karnataka) — Kudremukh, 100 per cent export unit, slurry pipeline to a port near Mangaluru; Maharashtra–Goa belt — Goa and Ratnagiri, exported through Marmagao.	5
20	Petroleum next major energy source after coal; found in anticlines and fault traps of tertiary rocks; Mumbai High, Gujarat (Ankeleshwar most important) and Assam (Digboi, Naharkatiya, Moran-Hugrijan — oldest producer); natural gas with petroleum — Mumbai High and allied fields, Cambay basin, Krishna-Godavari basin; HVJ pipeline (1,700 km) by GAIL; pipeline network expanded to 18,500 km; uses as CNG, PNG, fertilizer and power.	5
	OR: Rising prices and shortages of oil and gas; environmental problems of fossil fuels; nuclear (uranium and thorium in Jharkhand, Aravallis, monazite sands of Kerala); solar (photovoltaic); wind (Nagarcoil–Madurai, Jaisalmer); biogas (gobar gas); tidal (Gulf of Khambhat, Gulf of Kuchchh, Sunderbans); geothermal (Parvati valley — Manikaran, Puga valley — Ladakh).	5

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET B

Geography Chapter 5: **Minerals and Energy Resources** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** common salt, magnesium and bromine	Most minerals too diffused
2	**(c)** 10 kg	Ferro-manganese alloy
3	**(c)** Khetri mines in Rajasthan	India deficient in copper
4	**(c)** Panchpatmali in Koraput	Aluminium
5	**(a)** Both A and R are true and R is the correct explanation of A	Location factor
6	**(d)** mica	Splits into thin sheets
7	**(b)** limestone	Also for smelting iron
8	**(b)** Manikaran (Parvati valley) and Puga valley	HP and Ladakh
9	**(a)** GAIL (India)	1,700 km
10	**(d)** electricity	Wide range of applications

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	An accumulation of any mineral mixed with other elements, with sufficient concentration of mineral to make extraction commercially viable.	2
12	Refineries act as a nodal industry for synthetic textile, fertiliser and numerous chemical industries by providing raw materials.	2
13	Jharia, Raniganj, Bokaro (any two).	2
14	Electricity generated from oceanic tides using floodgate dams across inlets; Gulf of Khambhat, Gulf of Kuchchh or Gangetic delta in Sunderbans.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sedimentary — in beds or layers formed by deposition, accumulation and concentration in horizontal strata (coal, some iron ore); gypsum, potash salt and sodium salt formed by evaporation in arid regions. Residual — decomposition of surface rocks and removal of soluble constituents leaves a residual mass of weathered material containing ores like bauxite.	1½+1½
16	Clay-like substance from which alumina and then aluminium is obtained; formed by decomposition of rocks rich in aluminium silicates; deposits in Amarkantak plateau, Maikal hills, Bilaspur-Katni plateau; Odisha largest producer (Panchpatmali, Koraput); aluminium combines strength with lightness, conductivity and malleability.	3
17	Used in manufacturing steel and ferro-manganese alloy (about 10 kg per tonne of steel), bleaching powder, insecticides and paints; production 2018–19 — Madhya Pradesh (33 per cent), Maharashtra (27 per cent), Odisha (16 per cent), Karnataka (12 per cent), Andhra Pradesh (10 per cent).	3
	OR: Energy obtained by altering the structure of atoms, releasing heat used to generate electricity; uranium and thorium available in Jharkhand and the Aravalli ranges of Rajasthan; monazite sands of Kerala are rich in thorium.	1+2
18	Use public transport instead of individual vehicles; switch off electricity when not in use; use power-saving devices; use non-conventional sources; 'energy saved is energy produced'.	3
	OR: Peat — decaying plants in swamps, low carbon, high moisture, low heating capacity; lignite — low-grade brown coal, soft, high moisture (Neyveli); bituminous — buried deep, increased temperature, most popular; metallurgical — high-grade bituminous for blast furnaces; anthracite — highest quality hard coal.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	Everything from a pin to a building, ship, railway lines, roads, vehicles made of minerals; food contains minerals; toothpaste (silica, limestone, fluorite, titanium oxide, mica). Modes — veins and lodes in igneous/metamorphic rocks; beds or layers in sedimentary rocks; residual mass (bauxite); placer deposits (gold, silver, tin, platinum); ocean waters (salt, magnesium, bromine) and manganese nodules on ocean beds.	2+3
	OR: Nuclear — uranium and thorium in Jharkhand and Aravallis, thorium in Kerala's monazite; Solar — tropical, photovoltaic; Wind — Nagarcoil–Madurai cluster, AP, Karnataka, Gujarat, Kerala, Maharashtra, Lakshadweep, Jaisalmer; Biogas — rural areas, gobar gas; Tidal — Gulf of Khambhat, Kuchchh, Sunderbans; Geothermal — Manikaran (HP) and Puga valley (Ladakh).	5
20	Workable deposits only one per cent of crust; slow geological formation; rapid consumption; increasing costs and decreasing quality with depth; need planned and sustainable use; improved technology for low-grade ores at low cost; recycling metals; scrap metals; substitutes.	5
	OR: Hydroelectricity — fast-flowing water, renewable; multi-purpose projects like Bhakra Nangal, Damodar Valley Corporation, Kopili Hydel Project. Thermal — burning coal, petroleum and natural gas; non-renewable fossil fuels; per capita electricity consumption is an index of development; nuclear power stations; solar and wind plants.	5

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND